

SIMATS ENGINEERING

ASSESSMENT TOOL 2

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COURSE NAME: Computer Architecture

COURSE CODE: CSA1202

QUESTIONS:5

Total Marks:25

ANNEXURE A

CONCEPT MAPPING

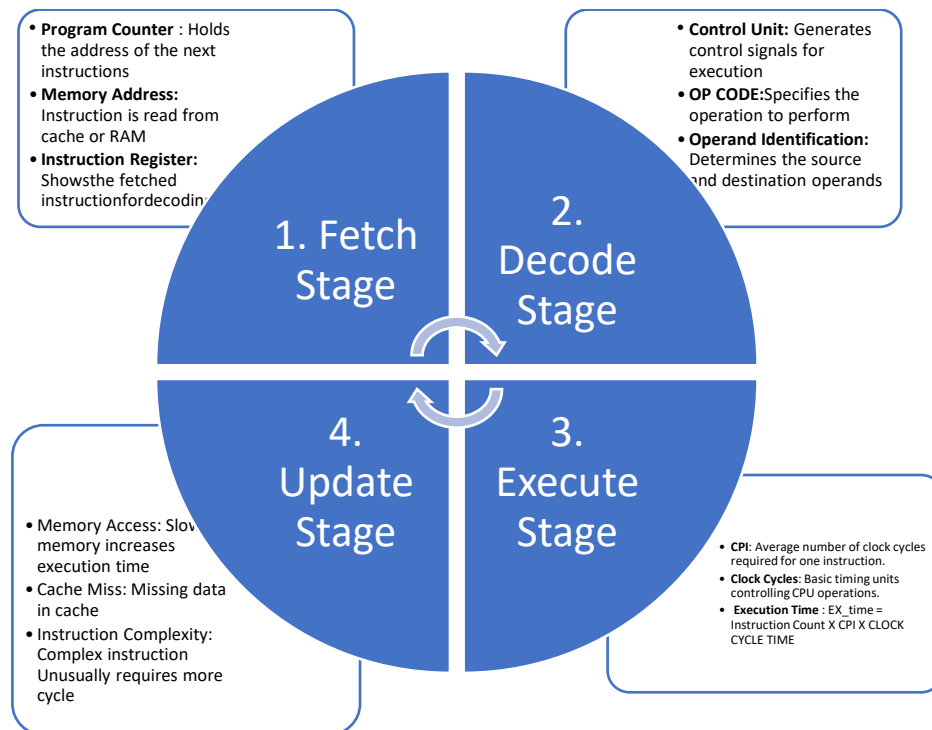
Code of Conduct:

I Lochana Sri R.S./ 192521348 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Lochana Sri R.S.

Signature of the student

1. Develop a comprehensive concept map linking the instruction execution cycle (fetch, decode, execute) with CPU performance parameters such as CPI, clock cycles, and execution time. Show how each stage contributes to overall execution time, and include connections to factors like memory access, instruction complexity, and pipeline stalls. Also, represent how benchmarking is used to evaluate performance.



2. Create a concept map comparing 8085 and 8086 architectures, including their instruction sets, register organization, addressing modes, and memory segmentation (in 8086). Clearly show how these architectural features influence program execution, multitasking capability, and efficiency in real-time applications.

8085 - MICROPROCESSOR:

An 8-bit microprocessor designed for simple embedded and control applications

Data bus : 8-bit , Address bus: 16-bit

Memory: Can access upto 64 KB

Registers: A, B, C, D, E , H, L
uses general purposes and accumulator registers.

Addressing Modes: Immediate, Direct, Register , Register Indirect

Instruction Set : Smaller Instruction set with fewer operations

8086 - MICROPROCESSOR:

A 16-bit processor with advanced architecture for faster execution

Data bus : 8-bit , Address bus: 20-bit

Memory: Can access upto 1 MB

Registers: AX, BX, CX, DX, SI , DI, BP, SP
Provides specialized registers for efficient processing.

Addressing Modes: Immediate, Direct, Indexed, Based, Based Indexed.

Instruction Set : Larger and more powerful instruction set.